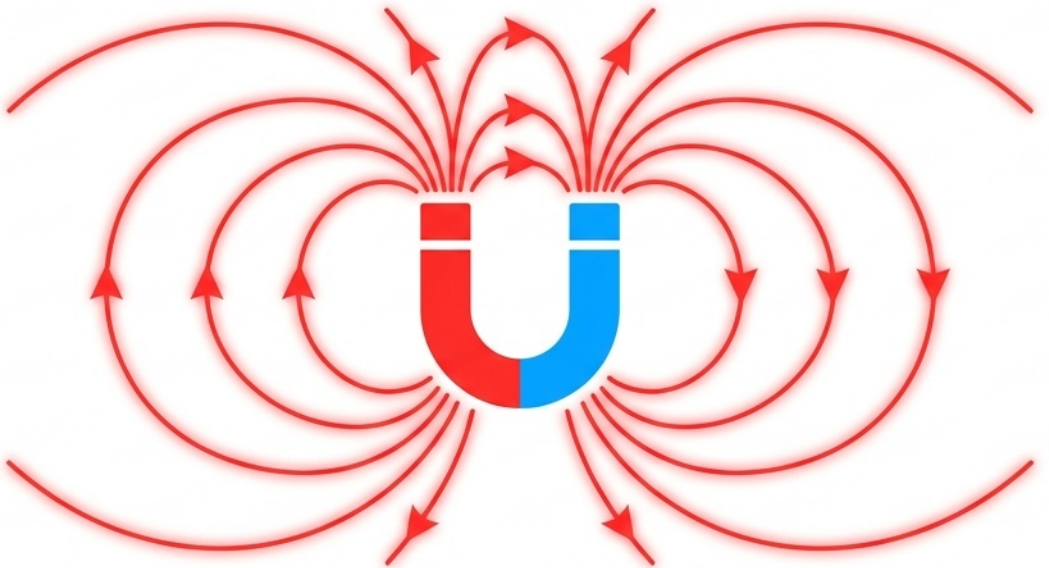
GCSE Physics

Lesson 7: Magnets and Electromagnets

Teacher: Dan (1 Practical)



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Starter: What do we already know?

1

Question: Name four magnetic elements.

Hint: Ferromagnetic metals

Answer: Iron, Steel, Cobalt, Nickel

2

Question: What is the direction of magnetic field lines outside a bar magnet?

Hint: North to South

Answer: From North pole to South pole (N → S)

3

Question: What is a solenoid?

Hint: Coiled wire

Answer: A cylindrical coil of wire carrying an electric current

4

Question: Give two advantages of an electromagnet over a permanent magnet.

Hint: Controllability

Answer: Can be switched on/off and strength can be varied

5

Question: Why is soft iron used as the core of an electromagnet instead of steel?

Hint: Loses magnetism when off

Answer: Iron easily magnetises and demagnetises (soft magnetic material)

6

Question: What happens to the strength of an electromagnet if coil turns are doubled?

Hint: More magnetic field loops

Answer: Strength increases (picks up more clips)

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Two Types of Magnet

Permanent Magnets

- **Always magnetic:** Produce their own constant magnetic field.
- **Poles:** Always have a North (N) and South (S) pole.
- **Interaction:** Can attract AND repel other magnets.

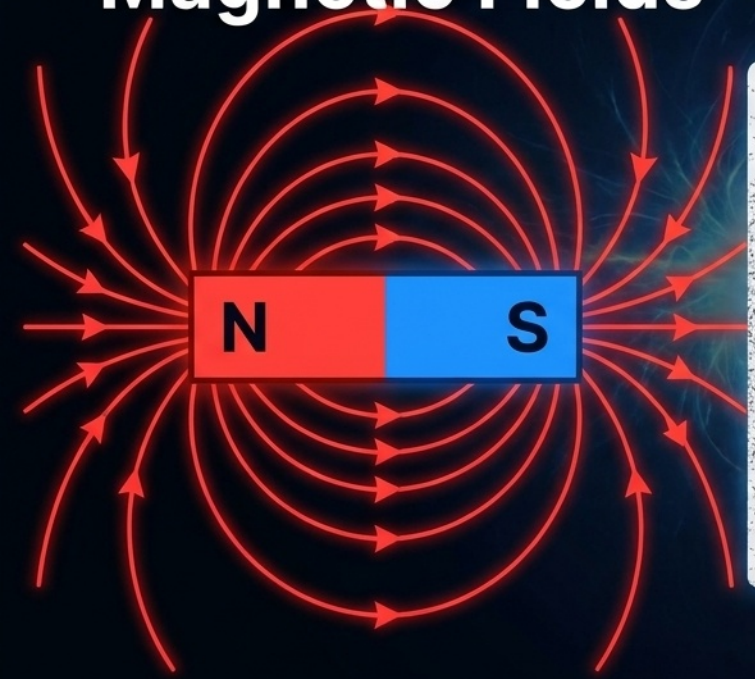
Induced Magnets

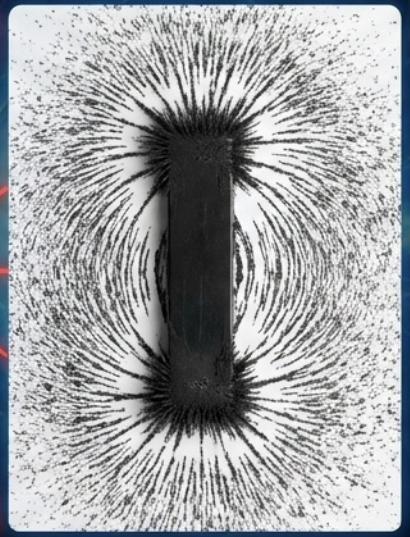
- **Temporary:** Become magnetic only when placed in a magnetic field.
- **Interaction:** Only ever experience a force of **attraction**.
- **Loss of magnetism:** Lose it quickly when removed from the field.

The Magnetic Four:
Iron, Steel, Cobalt, Nickel.

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Visualising the Invisible: Magnetic Fields





Key Rules

- **Direction:** Always travel from North to South.
- **Strength:** Indicated by line density (closer lines = stronger field).
- **Poles:** The field is strongest exactly at the poles.

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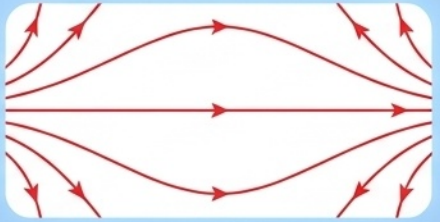
Quick Check: Pair & Share

Discuss and answer with the person next to you.

60 SECONDS

Question: A mystery metal is attracted to a magnet, but it cannot be repelled. Is it a permanent or induced magnet? Explain.

Animated Answer: Induced. Induced magnets only attract; permanent magnets can repel.



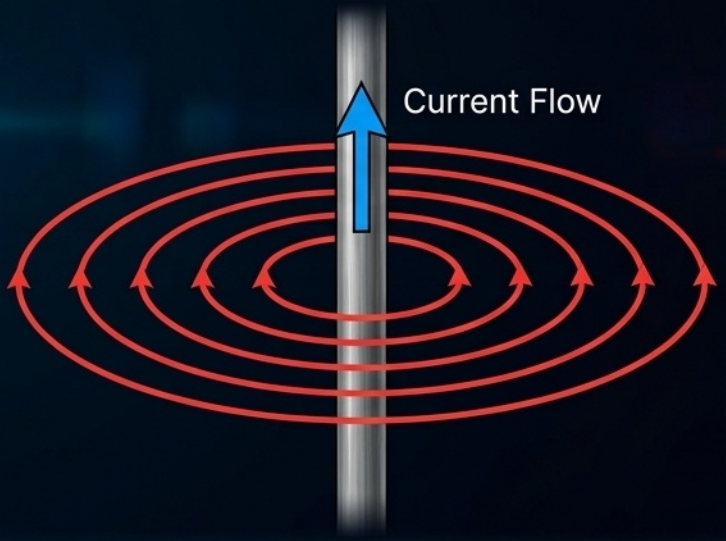
Question: Look at this diagram. Is the magnetic force strong or weak here?

Animated Answer: Weak. The field lines are far apart (low density).

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Electricity Creates Magnetism

- **The Effect:** An electric current flowing through a wire creates a magnetic field.
- **The Shape:** Concentric circles spreading outwards from the wire.



Right-Hand Grip Rule



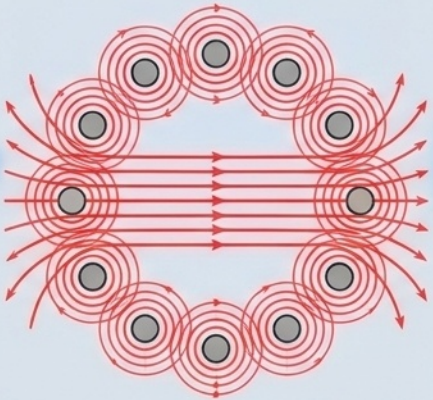
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The Solenoid: Supercharging the Field

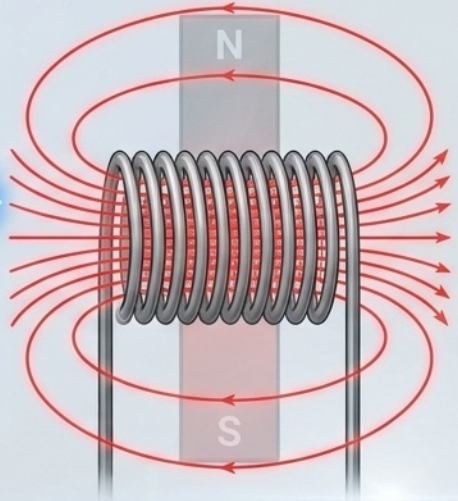
Coil the wire to create a **solenoid**.



The circular fields overlap and combine inside the coil.



The result? A strong, uniform magnetic field identical to a **bar magnet**.

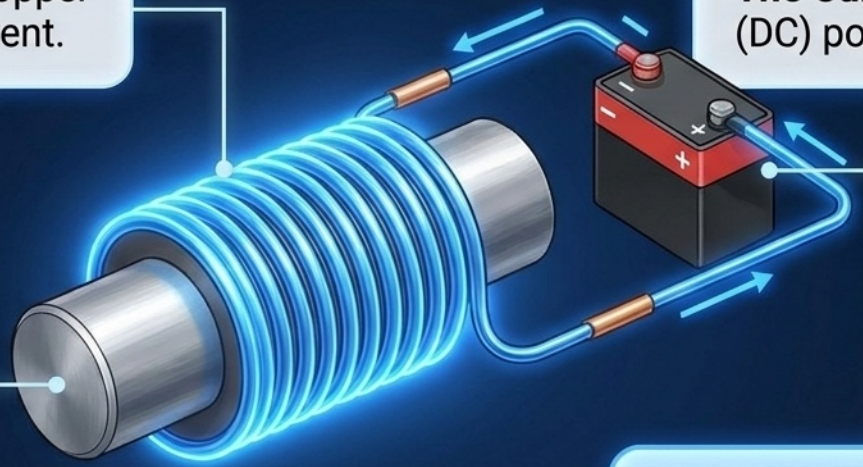


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Building an Electromagnet

The Coil: Insulated copper wire carrying the current.

The Current: Direct current (DC) power supply.



The Core: Soft Iron. Must be iron, not steel. Iron loses magnetism instantly when switched off.

How to make it stronger:

- **Increase** current.
- **Increase** number of coil turns.
- **Add** a soft iron core.

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Practical 1: Making & Testing an Electromagnet

Equipment

Iron nail, insulated copper wire, variable DC power supply, steel paper clips.

Prompt

How do the number of coil turns affect electromagnet strength?

Method (Step-by-Step):

1. Wind insulated wire tightly around the iron nail.
2. Start with **10 turns**. Connect to DC supply.
3. Set current to a safe fixed value (e.g., 2A).
4. Count the number of steel clips lifted.
5. Repeat for 20, 30, 40, and 50 turns. Keep current constant!

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Practical 1: Results & Conclusion

Number of Turns	Number of Clips Lifted
10	2
30	7
50	12

The Science:

- **Conclusion:** More turns = stronger electromagnet = more clips lifted.
- **Why?** More coils mean more overlapping magnetic field lines, increasing overall field density.

⚠ **Caution:** Wire gets hot! Switch off the DC supply between tests.

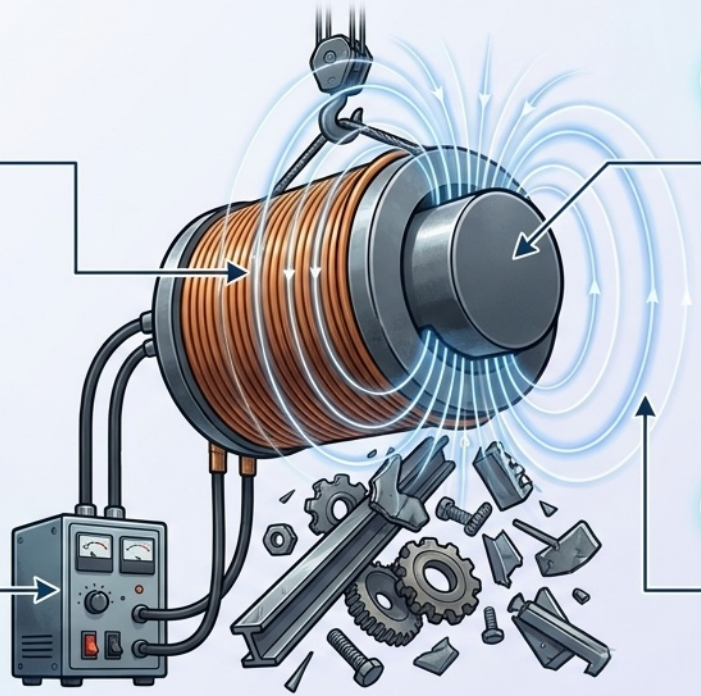
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Quick Check: Labelling Race

First pair to correctly identify all 4 labels wins!

1 Solenoid / Coil

2 Soft Iron Core



3 DC Power Supply

4 Magnetic Field Lines

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Real-World Applications

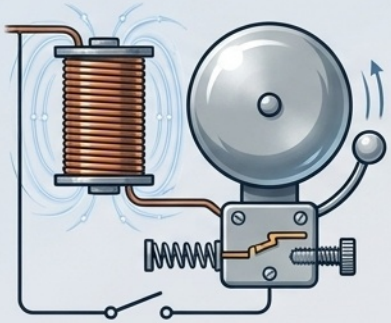
The Advantages: We can **switch them off**, and we can **change their strength**.

Scrap Metal Cranes



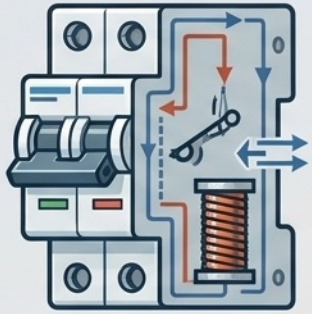
Turn on to lift heavy steel. Turn off to instantly drop the load.

Electric Bells



Magnet attracts a striker, breaking the circuit. The circuit resets, repeating the strike rapidly.

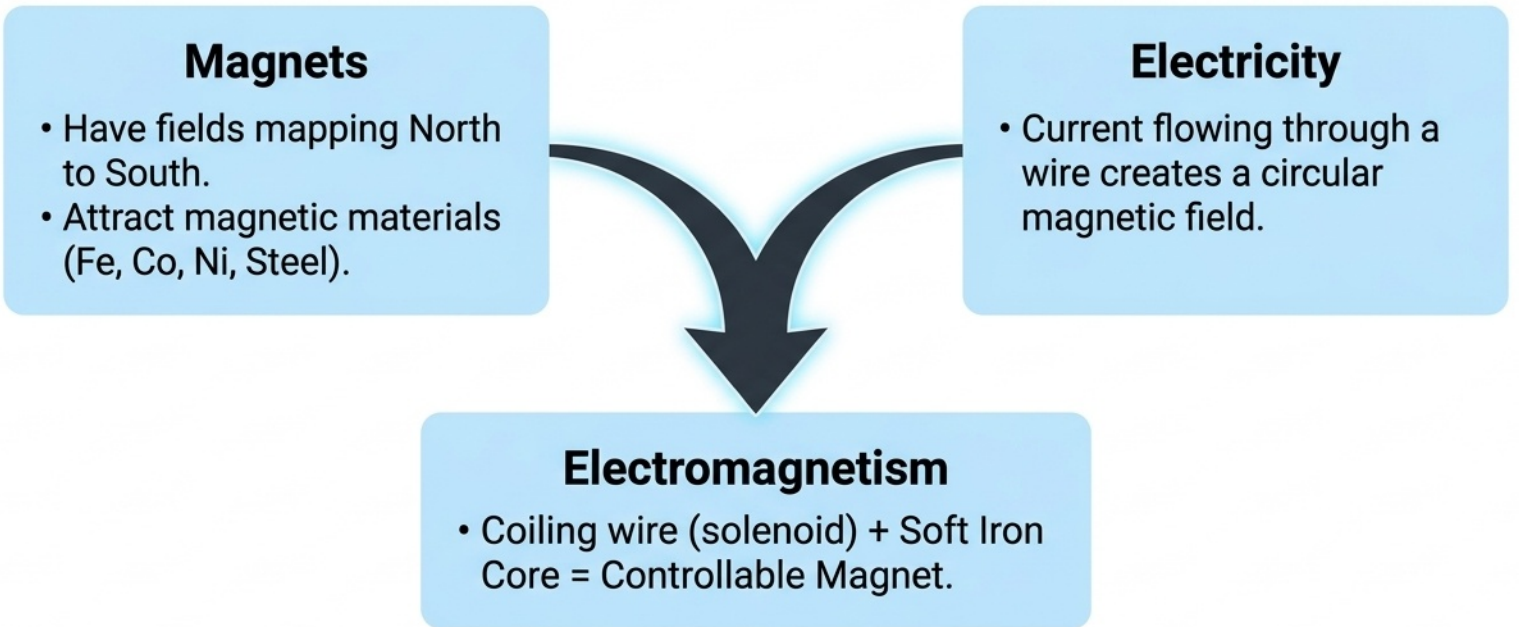
Circuit Breakers & Relays



High currents trigger a strong magnetic field. The field physically pulls a switch open to save a circuit.

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Lesson Summary: The Big Picture



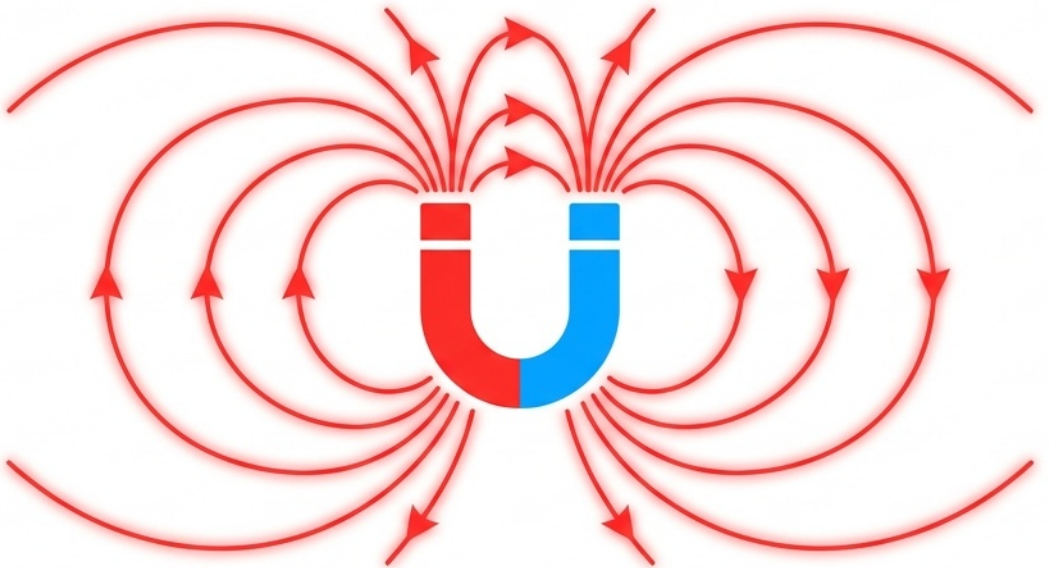
Electromagnetism bridges the gap between electrical energy and physical force.

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GCSE Physics

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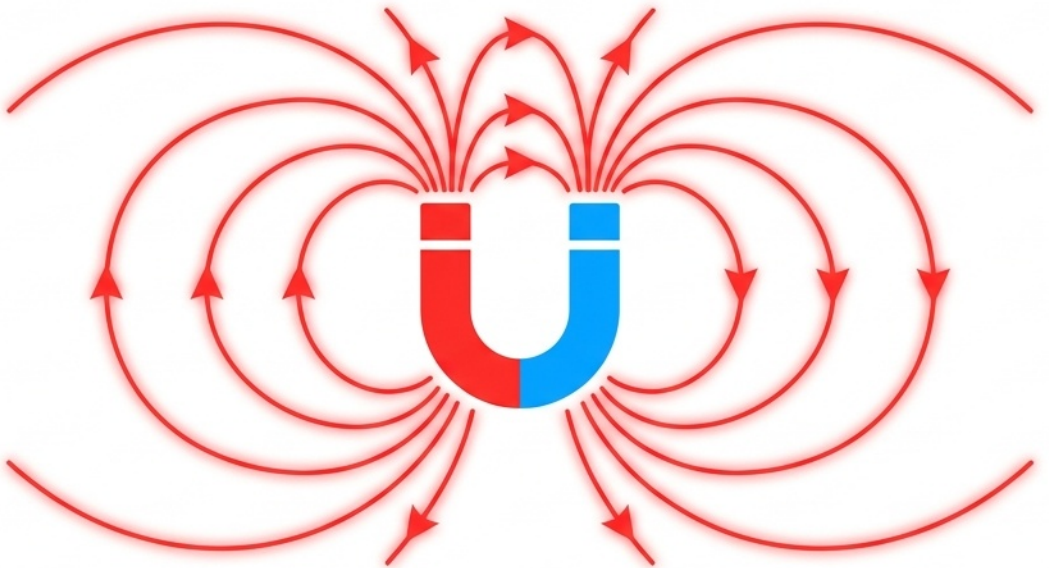


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